MTH422 : Bayesian method for causal inference in spatially-correlated multivariate time series

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1 Introduction

Assessing the causal impact of advertising campaigns on sales is a crucial task for advertising firms. However, challenges emerge when campaigns are conducted across multiple stores that are spatially correlated, and when sales data exhibit a low signal-to-noise ratio, making it difficult to identify advertising effects. The paper we chose for studying introduces a new Bayesian approach designed to tackle both issues. The method employs a multivariate structural time series model to account for spatial correlations between stores by applying a G-Wishart prior on the precision matrix.

The approach involves comparing two posterior distributions of a latent variable—one derived from the actual observed data in the test stores, and the other from estimated counterfactual outcomes. These counterfactuals are constructed using synthetic controls, which are linear combinations of sales data from multiple control stores during the causal period. Control stores are selected through an adapted Expectation-Maximization Variable Selection (EMVS) method.

To estimate the model parameters, a two-stage algorithm is introduced. To prevent overly wide prediction intervals, the model incorporates a stationarity constraint on the local linear trend, enforced via a recently developed prior. A comprehensive simulation study highlights the method's ability to detect subtle causal effects.

2 Notation and Setup

We now introduce the notations used throughout the analysis. Let n be the total number of **test stores** (which received the advertisement), and p_i be the number of **control stores** (which did not receive the advertisement) corresponding to test store i for $i = 1, \ldots, n$. The total number of control stores is $p = \sum_{i=1}^{n} p_i$. Each store has time series data of total length $T + P$, where $1, \ldots, T$ denotes the **pre-period** (before the advertising campaign), and $T + 1, \ldots, T + P$ denotes the **post-period** (during and after the advertising campaign). Let $w_t = (w_{1t}, \ldots, w_{n+p,t})^\top$ denote a binary treatment vector at time t, where each entry indicates whether a store received the advertisement. Since treatment assignment is fixed over time, we write $w_t = w$. Denote y_{it}^{obs} as the **observed sales** of store i at time t and y_{it}^{cf} as the **counterfactual sales**—what would have happened without the ad, which is usually unobserved for treated stores in the post-period. Let $\mathbf{Y}^{\text{obs}} = (y_{1t}^{\text{obs}}, \ldots, y_{nt}^{\text{obs}})^\top$ be the vector of observed outcomes for all test stores and $\mathbf{Y}^{\text{cf}} = (y_{1t}^{\text{cf}}, \ldots, y_{nt}^{\text{cf}})^\top$ be the vector of counterfactual outcomes. We define $\mathbf{Y}_{T+1:T+P}^{\text{obs}} = (Y_{T+1}^{\text{obs}}, \ldots, Y_{T+P}^{\text{obs}})$ as the observed data in the post-period, representing real sales after the advertisement, and $\mathbf{Y}_{T+1:T+P}^{\text{cf}} = (Y_{T+1}^{\text{cf}}, \ldots, Y_{T+P}^{\text{cf}})$ as the counterfactual post-period data, which are the estimated sales had the advertisement not been shown. The difference between $\mathbf{Y}^{\text{obs}}$ and $\mathbf{Y}^{\text{cf}}$ indicates the impact of the advertisement on sales.

We begin by introducing the common causal estimand used in this framework. For the i-th test unit, the causal effect is defined as the average difference between observed and counterfactual outcomes over the post-treatment period:

$$\frac{1}{P}\sum_{t=T+1}^{T+P}\left(y_{it}^{\text{obs}} - y_{it}^{\text{cf}}\right),$$

which corresponds to the average treatment effect across P time points, following the formulation in [1]. Extending this to n treated units, the aggregate treatment effect is expressed as:

$$\frac{1}{P}\sum_{t=T+1}^{T+P}\left(\mathbf{Y}_t^{\text{obs}} - \mathbf{Y}_t^{\text{cf}}\right). \tag{2.1}$$

3 Model and Prior

3.1 Model

Available Data and What we Want to Estimate By Model

We have access to the observed data $(Y_{1:T}, X_{1:T}, Y_{T+1:T+P}^{\text{obs}}, X_{T+1:T+P})$, where $Y_{1:T+P}$ represents the response (sales) from the test store and $X_{1:T+P}$ denotes the covariates (sales from control stores). Note that for Y after time point T, i.e., from $T+1$ to $T+P$, we only have observed data (sales after the ad) and not the counterfactual sales, which we aim to estimate using $X_{T+1:T+P}$.

We consider a multivariate structural time series model given by (to simplify the notation, we use Y_t instead of Y_t^{obs} in the current model) and the following setup:

$$Y_t = \mu_t + \delta_t + X_t\beta + \epsilon_t, \tag{1}$$

where Y_t, μ_t, δ_t, and ϵ_t are $n \times 1$ vectors standing for the response variable, trend, seasonality and measurement error respectively. n is the number of test stores, X_t is an $n \times p$ matrix containing data from p control stores at time t and β is a sparse $p \times 1$ vector of regression coefficients, where p can be very large. We allow each response in Y_t to have different number of control stores, and write

$$X_t = \begin{pmatrix} x_{11,t} & \cdots & x_{1p_1,t} & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \cdots & 0 & x_{21,t} & \cdots & x_{2p_2,t} & \cdots & 0 \\ \vdots & & \vdots & \vdots & & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \cdots & x_{np_n,t} \end{pmatrix}$$

with $\sum_{i=1}^{n} p_i = p$. ϵ_t is an independent and identically distributed (i.i.d.) error process.

The trend of the time series is modeled as:

$$\mu_{t+1} = \mu_t + \tau_t + u_t, \tag{2}$$

The process τ_t can be modeled as a stationary VAR(1) process, or in vector form as:

$$\tau_{t+1} = D + (\tau_t - D) + \upsilon_t \tag{3}$$

where D is an $n \times 1$ vector and Φ is an $n \times n$ matrix of the coefficients of the VAR(1) process with eigenvalues having modulus less than 1. If no stationarity restriction is imposed on τ_t, we model it by

$$\tau_{t+1} = \tau_t + \upsilon_t, \tag{4}$$

where υ_t is an i.i.d. error process.

The seasonal component δ_t in (1) is assumed to follow the evolution equation:

$$\delta_{t+1} = -\sum_{j=0}^{s-2} \delta_{t-j} + w_t, \tag{5}$$

We assume that the residuals of (1)–(5) are mutually independent and time invariant, and are distributed as multivariate normals with mean $\mathbf{0}_{n\times 1}$ and covariance matrices $\mathbf{\Sigma}_{\eta_t}$, $\mathbf{\Sigma}_{\epsilon_t}$, and $\mathbf{\Sigma}_{w_t}$, respectively.

Let $\alpha_t = (\mu_t', \tau_t', \delta_t', \cdots, \delta_{t-S+2}')'$ and $\eta_t = (\mu_t', v_t', w_t')'$, we can be represented as a linear Gaussian state-space model as below :

$$Y_t = z\alpha_t + X_t\beta + \epsilon_t \tag{6}$$

$$\alpha_{t+1} = c + T\alpha_t + R\eta_t \tag{7}$$

where z, c, T, and R can be rearranged accordingly based on the model (1)–(5) and $\epsilon_t \sim \mathcal{N}(0, \Sigma_{\epsilon_t})$, $\eta_t \sim \mathcal{N}(0, Q)$, $Q = \text{bdiag}(\Sigma_{\mu_t}, \Sigma_{v_t}, \Sigma_{w_t})$ are mutually independent here bdiag refers to a block-diagonal matrix with entries as specified. If τ_t is a nonstationary process in (3), then we set $c = 0$.

3.2 Priors for Parameter

We will now discuss the priors for the parameters in the model. We separate the parameters into four blocks: the time varying parameter α_t, the stationarity constraint parameters D and β, the covariance matrices of the error terms $\mathbf{\Sigma}_{\eta_t}$, $\mathbf{\Sigma}_{\epsilon_t}$, and $\mathbf{\Sigma}_{w_t}$, and the sparse regression parameter β.

For the time varying parameter, we give a prior $\alpha_1 \sim \mathcal{N}(a, P)$ with a is the mean and P is the covariance matrix. For the covariance matrices of the errors, we choose

$$\mathbf{\Sigma}_{\eta_t}^{-1} \sim \mathcal{W}_G(\nu, H), \qquad \mathbf{\Sigma}_{\epsilon_t}^{-1} \sim \mathcal{W}_G(\nu, k_2^2(n+1)H),$$
$$\mathbf{\Sigma}_{w_t}^{-1} \sim \mathcal{W}_G(\nu, k_3^2(n+1)H),$$

where $\mathcal{W}_G$ stands for a G-Wishart distribution. For the stationarity constraint parameter D, we choose a conjugate prior $D \sim \mathcal{N}(0, I_n)$.

4 Posterior Computation by 2 Stages

Using the available data, in Stage 1 we apply the EMVS (Expectation-Maximization for Variable Selection) method to estimate the sparse regression coefficients $\hat{\beta}$. This method is chosen due to its computational efficiency and scalability for large datasets. Once $\hat{\beta}$ is obtained, we compute the residuals $\tilde{Y}_t = Y_t - X_t\hat{\beta}$, which represent the de-trended series by removing the regression effect. These residuals $\tilde{Y}_t$ are then modeled using a Bayesian state-space model to capture the underlying trend and noise.

In Stage 2, we apply MCMC methods to draw samples from our parameters.

1. **Stage 1: EMVS Step.** Using the EMVS algorithm, we find the optimized value for $\hat{\beta}$. After obtaining $\hat{\beta}$, we plug it into Equation (6) and subtract $X_t\hat{\beta}$ from Y_t. We denote the resulting data as $\tilde{Y}_t$, and work with the following state-space model:

$$\tilde{Y}_t = Z\alpha_t + \epsilon_t, \tag{8}$$

$$\alpha_{t+1} = c + T\alpha_t + R\eta_t. \tag{9}$$

2. **Stage 2: MCMC Step.** In the MCMC step, we sample the parameters from their corresponding posterior distributions. These parameters include the time-varying states $\alpha_{1:T}$, the stationary constraint parameters $\boldsymbol{D}$ and $\boldsymbol{\Phi}$, and the covariance matrices of the residual noise, with $\Sigma_\eta^{-1} \sim$ Wishart and $\Sigma_\epsilon^{-1} \sim$ Wishart.

   The details of the algorithm are presented in the supplementary material (Ning et al., 2018).

The proposed two-stage estimation algorithm is summarized as follows:

Stage 1: Choose initial values for $\beta^{(0)}$, $\alpha_{1:T}^{(0)}$, and $P_{1:T}^{(0)}$ using the revised EMVS algorithm to obtain the optimized estimate $\hat{\beta}$.

Stage 2: Given $\tilde{Y}_t$, sample the parameters using MCMC with the following sub-steps:

(a) Generate $\alpha_{1:T}$ using the Kalman filter and simulation smoother method.

(b) Generate $\boldsymbol{\Phi}$ using the Metropolis-Hastings algorithm.

(c) Generate $\boldsymbol{D}$.

(d) Generate the covariance matrices Σ_η and Σ_ϵ from their respective G-Wishart posterior distributions.

(e) Repeat from Step (i) until convergence.

Note: Skip Steps (ii) and (iii) if no stationarity restriction is imposed on τ.

5 Method for Causal Estimand

Before proceeding with our proposed causal estimand, we define the one-sided Kolmogorov-Smirnov (KS) distance between two probability distributions.

The one-sided KS distance quantifies the maximum difference between two cumulative distribution functions (CDFs) in one direction. It is defined as:

$$D^+ = \sup_x [F_1(x) - F_2(x)], \tag{10}$$

where $F_1(x)$ and $F_2(x)$ are two CDFs, and $\sup_x$ denotes the supremum (least upper bound) over all values of x. This distance varies from 0 to 1 and measures the maximum positive difference between the first and second CDFs.

We now present a novel approach for quantifying causal effects in retail settings where test stores exhibit spatial correlation. The core idea is: instead of relying on point estimates, we leverage the entire posterior distributions of our model's latent variables under both observed and counterfactual scenarios. To compare these distributions meaningfully, we employ the one-sided KS distance.

This approach offers several advantages:

- It captures the full uncertainty in causal estimates rather than reducing it to a single number.
- It naturally accounts for spatial correlations between test stores, which traditional methods often overlook.
- It provides an intuitive framework for determining statistical significance via comparison with appropriate thresholds.

Detailed Approach:

Let x_{it} be the data for the synthetic control for the i-th test store at time t. Define $\mathbf{X}_{1:T+P} = (\mathbf{X}_1, \ldots, \mathbf{X}_{T+P})$, where each $\mathbf{X}_t$ is an $n \times p$ matrix of data from p control stores. Let μ_{it} be a latent variable of interest, and $\boldsymbol{\mu}_i = (\mu_{i1}, \ldots, \mu_{iT})$ be an $n \times 1$ vector.

Define the following posterior distributions:

$$p\left(\sum_{t=T+1}^{T+P} \mu_{it} \mid \mathbf{Y}^{\text{obs}}_{1:T+P}, \mathbf{X}_{1:T+P}\right), \tag{11}$$

$$p\left(\sum_{t=T+1}^{T+P} \mu_{it} \mid \mathbf{Y}^{\text{obs}}_{1:T}, \mathbf{Y}^{\text{cf}}_{i,T+1:T+P}, \mathbf{X}_{1:T+P}\right). \tag{12}$$

The proposed causal estimand for the i-th store is defined as the one-sided KS distance between the two distributions:

$$\sup_x \left[\mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid y^{\text{obs}}_{i,1:T}, y^{\text{cf}}_{i,T+1:T+P}, x_{1:T+P}\right) - \mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid y^{\text{obs}}_{1:T+P}, x_{1:T+P}\right)\right], \tag{13}$$

where $\mathcal{F}(\cdot)$ denotes the corresponding cumulative distribution function.

Since test stores are spatially correlated, the causal effect for the i-th store is formally defined as:

$$\sup_x \left[\mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid \mathbf{Y}^{\text{obs}}_{1:T}, \mathbf{Y}^{\text{cf}}_{i,T+1:T+P}, \mathbf{X}_{1:T+P}\right) - \mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid \mathbf{Y}^{\text{obs}}_{1:T+P}, \mathbf{X}_{1:T+P}\right)\right]. \tag{14}$$

A larger value of the one-sided KS distance implies a potentially larger scale of causal impact. An impact is declared statistically significant if the KS distance exceeds a threshold, which is computed based on datasets randomly drawn from the posterior predictive distribution.

6 Step-wise Breakdown of the Whole Approach

Step 1: Apply the two-stage algorithm to obtain posterior samples for model parameters using data from the period without causal impacts (i.e., the pre-treatment period).

Step 2: Using the posterior samples from Step 1, draw samples of $Y^{\text{cf}}_{T+1:T+P}$ from its predictive posterior distribution:

$$p\left(Y^{\text{cf}}_{T+1:T+P} \mid Y^{\text{obs}}_{1:T}, X_{1:T+P}\right).$$

Step 3: Generate k counterfactual datasets, denoted by $Y^{\text{cf}(j)}_{T+1:T+P}$ for $j \in \{1, \ldots, k\}$, by sampling from the predictive posterior distribution. For each j-th counterfactual dataset:

- Fit $Y^{\text{cf}(j)}_{T+1:T+P}$ into the model and obtain posterior samples of the latent trend (as in Equation (12)).
- Fit the observed data $Y^{\text{obs}}_{1:T+P}$ into the model and sample from its posterior distribution (as in Equation (11)).

Step 4: Use the one-sided Kolmogorov-Smirnov (KS) distance to quantify the difference between the posterior distributions of the trend under observed data and under the counterfactual datasets. Stack the MCMC samples from all k simulated counterfactual datasets to approximate the counterfactual distribution. For each store $i = 1, \ldots, n$, calculate:

$$\sup_x \left[\frac{1}{k} \sum_{j=1}^{k} \mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid Y_{1:T}^{\text{obs}}, Y_{i,T+1:T+P}^{\text{cf}(j)}, X_{1:T+P} \right) - \mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid Y_{1:T+P}^{\text{obs}}, X_{1:T+P} \right) \right], \tag{15}$$

where $\mathcal{F}(\cdot)$ denotes the empirical distribution function of the posterior samples.

Step 5: Compute the $k \times (k-1)$ pairwise one-sided KS distances between posterior distributions of the trend for each pair of counterfactual datasets:

$$\sup_x \left[\mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid Y_{1:T}^{\text{obs}}, Y_{i,T+1:T+P}^{\text{cf}(j)}, X_{1:T+P} \right) - \mathcal{F}\left(\sum_{t=T+1}^{T+P} \mu_{it} \leq x \mid Y_{1:T}^{\text{obs}}, Y_{i,T+1:T+P}^{\text{cf}(j')}, X_{1:T+P} \right) \right], \tag{16}$$

where $j, j' \in \{1, \ldots, k\}$ and $j \neq j'$.

Then, for each store i, determine the 95th upper percentile of these distances and use it as the threshold for significance. If the KS distance computed in Step 4 is smaller than this threshold, the corresponding causal impact is declared *not significant*.

7 Simulation

We generate five datasets with spatial correlation, and suppose the precision matrices in the model are based on the adjacency matrix as given below:

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

We posit that variables are organized in a linear sequence, each linked solely to its immediate neighbors. We construct daily time series over an arbitrary period from January 1, 2016, to April 9, 2016, with a shift commencing on March 21, 2016. We specify dates in the simulation to assist in outlining the intervention phase. Initially, we develop five multivariate datasets for test stores, varying in their level of influence, and designate them as Datasets 1–5.

For each Dataset i, where $i = 1, \ldots, 5$, the trend is modeled from $\mu_{i,t} \sim \mathcal{N}(0.8\mu_{i,t-1}, 0.1^2)$ with an initial value $\mu_{i,0} = 1$. The weekly variations are crafted from two sinusoids sharing the same frequency 7 as follows:

$$\delta_{i,t} = 0.1 \times \cos(2\pi t/7) + 0.1 \times \sin(2\pi t/7).$$

Further datasets for 10 control stores are generated, each following an AR(1) process with a coefficient of 0.6 and a standard deviation of 1. We assign the first and second datasets regression coefficients $\beta_1 = 1$, $\beta_2 = 2$, while setting the remainder to 0. We then produce the residuals ϵ_t in the observation equation using a multivariate normal distribution $\mathcal{N}(0, \Sigma)$,

where the precision matrix displays a sparse structure as defined in (6.1). We set the diagonal terms of Σ^{-1} to 10, and the non-zero off-diagonal terms to 5. The resulting data for test stores combine the simulated values of μ_t, δ_t, $X_t\beta$, and ϵ_t. The causal effects are introduced as follows: for each Dataset i, $i = 1, \ldots, 5$, we apply an impact scale $\frac{t-1}{2} \times (\log 1, \ldots, \log 20)$ from March 21, 2016, to April 9, 2016. Notably, no causal impact is incorporated into Dataset 1.

We incorporate the adjacency matrix structure, into both the observed and hidden processes within the model. Subsequently, we employ a two-stage approach to determine the parameters. In this simulation, the EMVS (Expectation-Maximization Variable Selection) method is employed as a straightforward tool to identify which control stores influence the test stores' sales. Starting with all influence values (β) set to zero, EMVS analyzes the sales data, including trends and seasonal patterns, guided by the adjacency matrix that highlights relationships between nearby stores. Through an iterative process, it adjusts these values, pinpointing significant impacts (e.g., $\beta_1 = 1$ and $\beta_2 = 2$) while dismissing negligible ones, using a threshold to distinguish them.

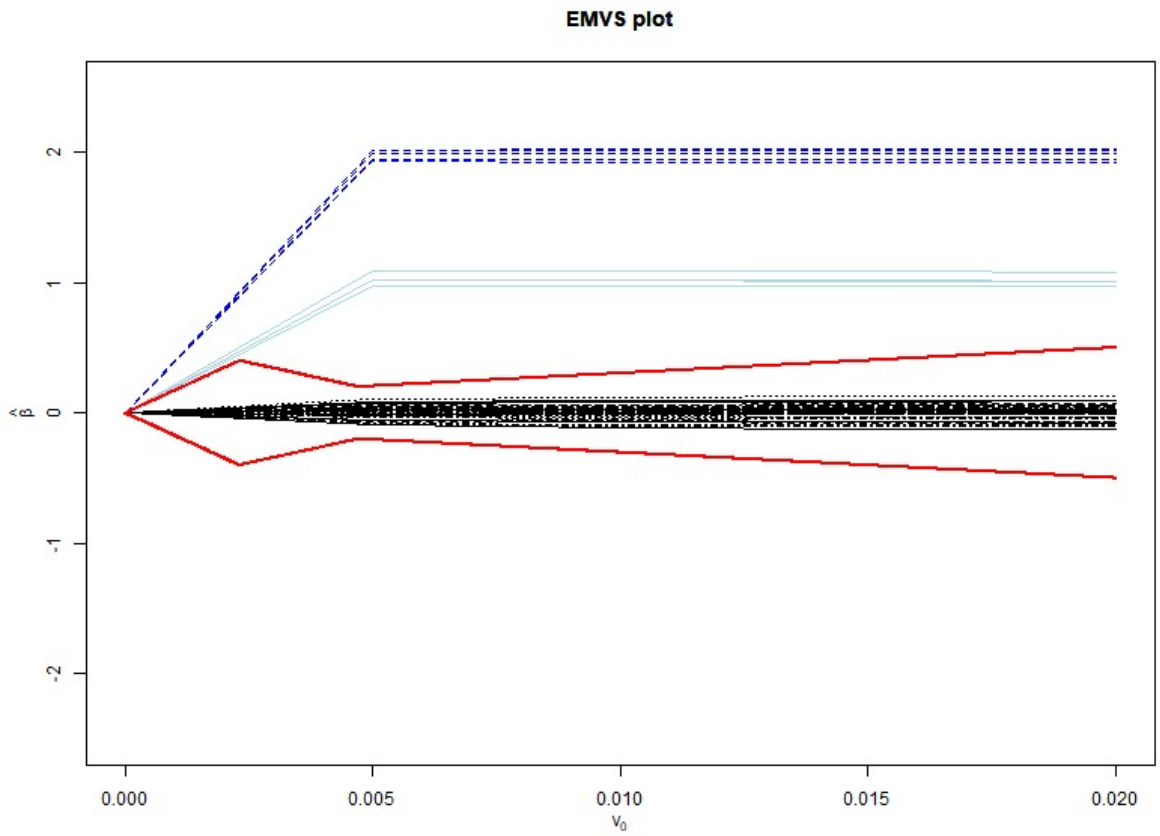

Figure 1: EMVS The dark blue lines represent parameters simulated with value 2, light blue with value 1, and black with value 0. The red lines indicate the estimated β_i^{th} bounds; parameters lying within these bounds are considered zero.

DAEMVS, or Deterministic Annealing Expectation-Maximization Variable Selection [2], is an enhanced version of the EMVS method used in your simulation to figure out which control stores affect the test stores' sales. Think of it like a smarter version of the EMVS assistant that starts with a clean slate (all influence values β set to zero) and looks at the sales data, including trends and seasonal patterns, guided by the adjacency matrix that shows how nearby stores are connected.

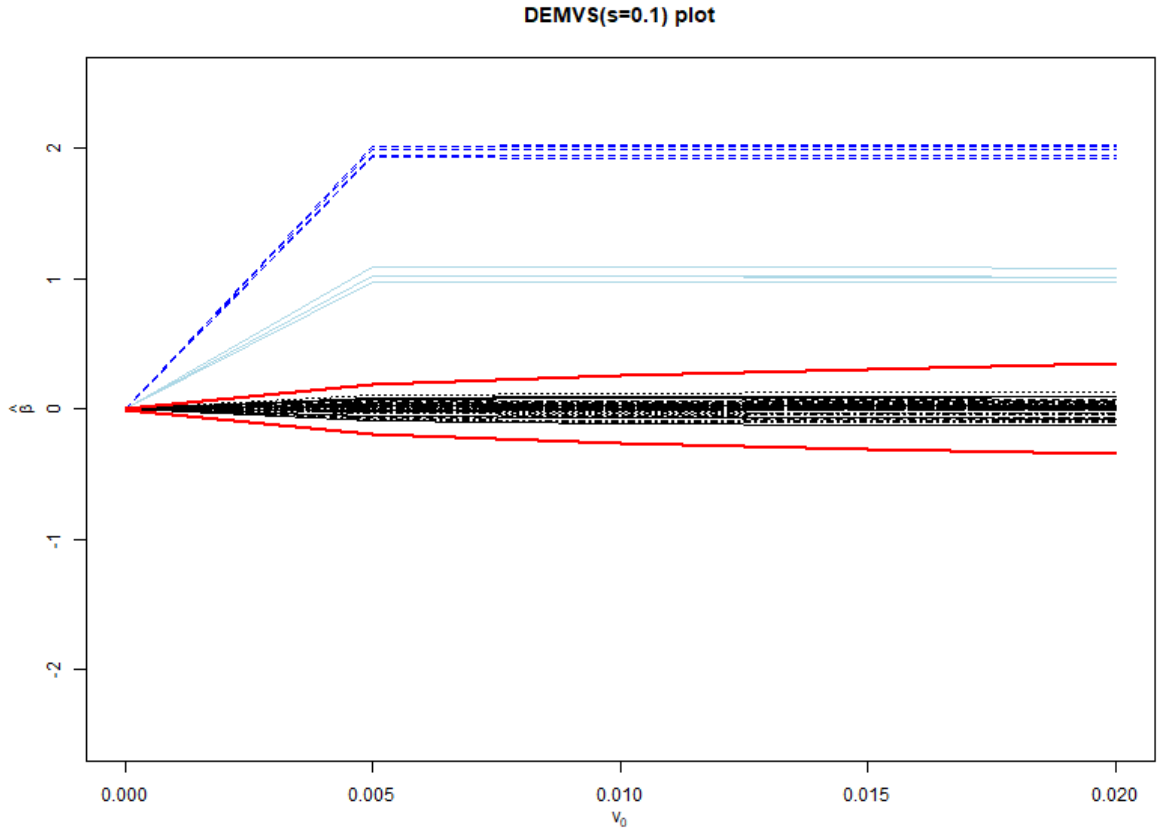

Figure 2: DAEMVS (s = 0.1) The dark blue lines represent parameters simulated with value 2, light blue with value 1, and black with value 0. The red lines indicate the estimated β_i^{th} bounds; parameters lying within these bounds are considered zero.

What makes DAEMVS special is its "temperature" trick: it starts with a high temperature (controlled by a parameter s, ranging from 0 to 1) to get a broad view of the data, which helps it avoid getting stuck on a wrong answer too early. As it lowers the temperature (moving s closer to 0), it zooms in on the best solution, adjusting the influence values step by step to highlight significant impacts (like $\beta_1 = 1$ and $\beta_2 = 2$) while ignoring the unimportant ones. This gradual approach helps DAEMVS find the most accurate overall picture of which control stores matter, making it better at avoiding mistakes and improving the results compared to the basic EMVS method.

We also consider using other two models: one ignores the stationarity constraint for $\boldsymbol{\tau_t}$ (henceforth the "nonstationary model"); another ignores the time dependency of the model (henceforth the "misspecified model"). To be more explicit, for the nonstationary model, we let the local linear trend follow eqn(4). The misspecified model is given by $\mathbf{Y}_t^* = \mathbf{X}_t^* \boldsymbol{\beta} + \varsigma_t$, with ς_ts are i.i.d random errors with multivariate normally distributed and mean $\mathbf{0}$ by ignoring their dependency. We conduct DAEMVS with s= 0.1 for both of the two models. Below plots are the results:

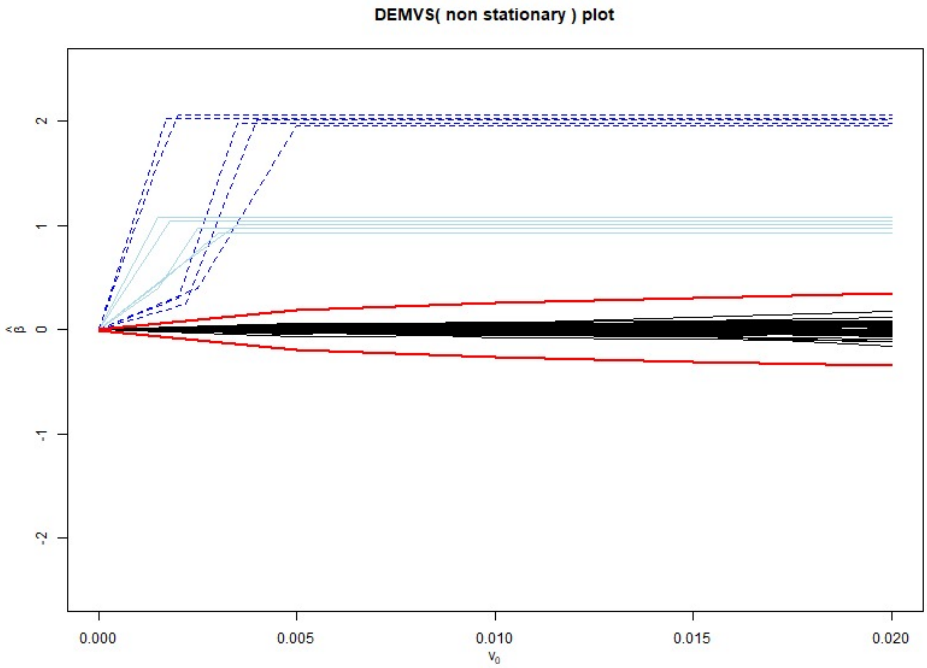

Figure 3: DAEMVS (with $s = 0.1$) estimation of β based on the simulated datasets using the nonstationary model. Dark blue lines indicate parameters with simulated value 2, light blue for value 1, and black for value 0. Red lines denote the calculated β_i^{th}; parameters within these are considered zero.

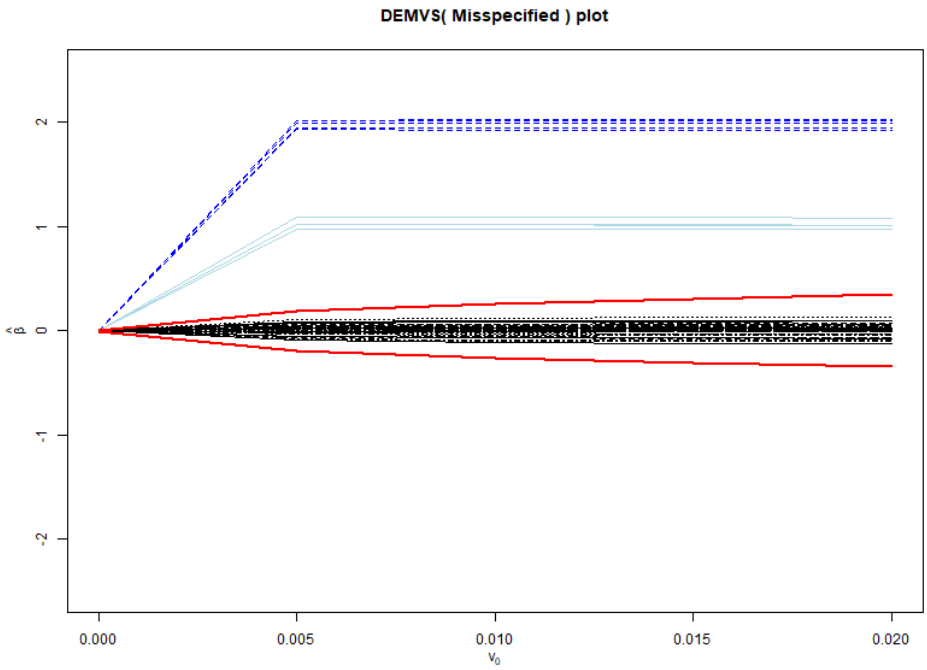

Figure 4: DAEMVS (with $s = 0.1$) estimation of β based on the simulated datasets using the misspecified model. Dark blue lines indicate parameters with simulated value 2, light blue for value 1, and black for value 0. Red lines denote the calculated β_i^{th}; parameters within these are considered zero.

In Stage 2, we substitute the estimated $\hat{\beta}$ and compute $\tilde{Y}_t$. The remaining parameters are assigned priors as follows: $\boldsymbol{\alpha}_1 \sim \mathcal{N}(0, I)$. If $\boldsymbol{\tau}_t$ exhibits nonstationarity, the initial state is treated as a diffuse random variable with high variance [3]. The covariance matrix of $\boldsymbol{\tau}_t$ is then set to $10^6 \times I_5$. We use $\nu = 1$ and set $k_1 = k_2 = k_3 = 0.1$. The matrix H is chosen as the identity matrix I_5, and priors are specified for 25 decomposed parameters. We specify the prior for $\boldsymbol{\Phi}$ as $\mathcal{N}(0, \sqrt{5}^2)$, and assume $\iota \sim$ Bernoulli(0.5). A total of 10,000 MCMC iterations are performed, with the first 2,000 samples discarded as burn-in.

7.1 Performance of the commonly use inference method

In this section, we assess the performance of a commonly used method for estimating causal effects. The estimation strategy is based on comparing the actual observed data during the period of causal influence with the model-predicted counterfactual outcomes—i.e., the values that would have been expected in the absence of any intervention.

The analysis proceeds in two stages. In Stage 1, we use the DAEMVS algorithm with a sparsity parameter set to $s = 0.1$ to estimate the regression coefficients $\hat{\beta}$ for the model specified in equations (6)–(7). To ensure temporal stability, we impose a stationarity constraint on the local linear trend component τ_t.

Stage 2 explores the effect of this stationarity assumption by comparing two model settings: one with a stationary trend and one without it. We use Dataset 4 to illustrate the results, which are visualized in figure below. This figure contains four subplots: the left two correspond to the model with a nonstationary local linear trend, and the right two represent the model where the trend is stationary. Before the intervention date—March 21, 2016—the two models yield similar posterior medians and 95% credible intervals obtained from the two models are close.

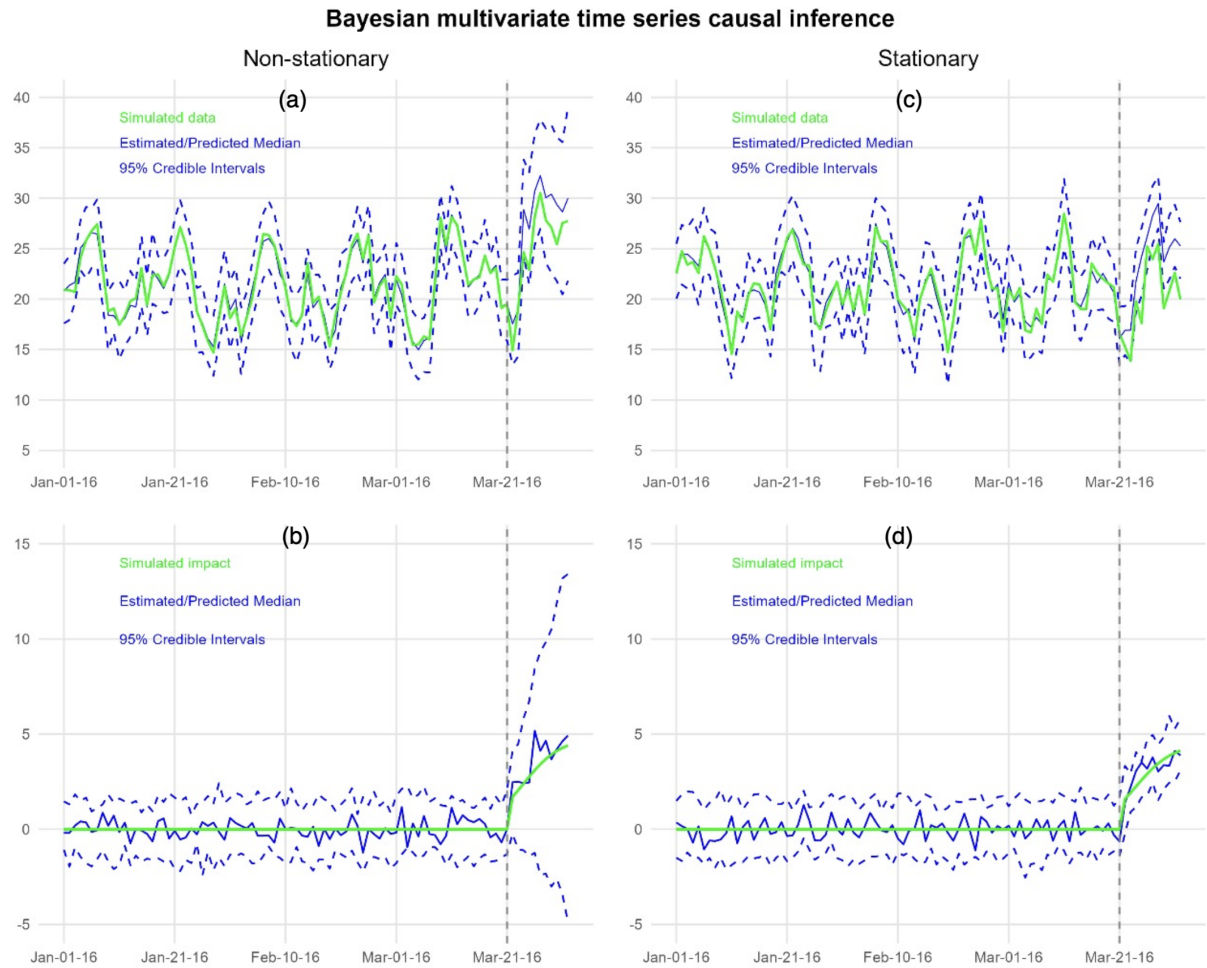

Figure 5 Plot of the causal impact in Dataset 4 using models with a stationary and a non-stationary local linear trend. (a) and (c) are the plots of estimation (before March 21, 2016) and prediction (after March 21, 2016) of Dataset 4 without stationarity constraint (left) and with stationarity constraint (right). The green line is the simulated dataset, the blue line is the estimated posterior median of the dataset using the model, the dashed blue line is the corresponding 95% credible and prediction intervals. (b) and (d) are the plots of estimated causal impact by taking the difference between the observed data and Bayesian estimates using the model with a nonstationary local linear trend (left) and the model with a stationary local linear trend (right). The green line is the simulated true impact, the blue line is the estimated median of the impact, the dashed blue lines are the corresponding 95% credible and prediction intervals.

In the case where the local linear trend is modeled as nonstationary, the resulting prediction intervals are considerably wider and grow more rapidly over time compared to those generated under a stationary trend assumption. As a consequence, during the intervention period, the observed data fall entirely within these expansive intervals, which prevents the model from identifying any significant causal effect. In contrast, the stationary trend model produces more moderately expanding prediction intervals, enabling it to distinguish deviations in the observed data and thus successfully detect the causal impact.

7.2 Performance of the KS distance method to infer causality

Now we evaluate the effectiveness of the proposed method using the same simulated dataset. The computation of the one-sided Kolmogorov-Smirnov (KS) distance is carried out as follows, for each $i = 1, \ldots, n$. Specifically, the one-sided KS distance is given by

$$\sup_x \left[\frac{1}{k} \sum_{j=1}^{k} \mathcal{F}\left(\sum_{t=T+1}^{T+m} \mu_{it} \leq x \middle| \mathbf{Y}_{1:T}^{\text{obs}}, \mathbf{Y}_{T+1:T+m}^{\text{cf}(j)}, \mathbf{X}_{1:T+m} \right) - \mathcal{F}\left(\sum_{t=T+1}^{T+m} \mu_{it} \leq x \middle| \mathbf{Y}_{1:T+m}^{\text{obs}}, \mathbf{X}_{1:T+m} \right) \right]$$

where m ranges from 1 to P. This allows us to observe the temporal evolution of the KS distances.

The resulting plots are displayed in below, where each subplot corresponds to a separate simulated dataset. In each subplot, the red line indicates the one-sided KS distances between a test store's posterior and that of its counterfactuals, while the light blue line marks the corresponding threshold. The threshold is computed using $k = 30$ simulated counterfactual datasets.

Among the datasets, only Dataset 1 shows one-sided KS distances consistently below the threshold, indicating the absence of any intervention effects. This highlights the method's capability to distinguish between impacted and non-impacted datasets. In Dataset 2, a minor early-period impact is observed, suggesting no significant causal effect during the initial three prediction intervals. However, the method successfully detects an effect beginning in the fourth prediction period.

A summary of the results is provided in Table 1. Compared to the commonly used method, the new approach demonstrates a clear improvement in identifying causal impacts. In Datasets 3 through 5, the one-sided KS distances consistently exceed their respective thresholds. Moreover, as the strength of the impact increases, the distances also grow larger. The thresholds rise over time as well, reflecting the widening prediction intervals for the trend components.

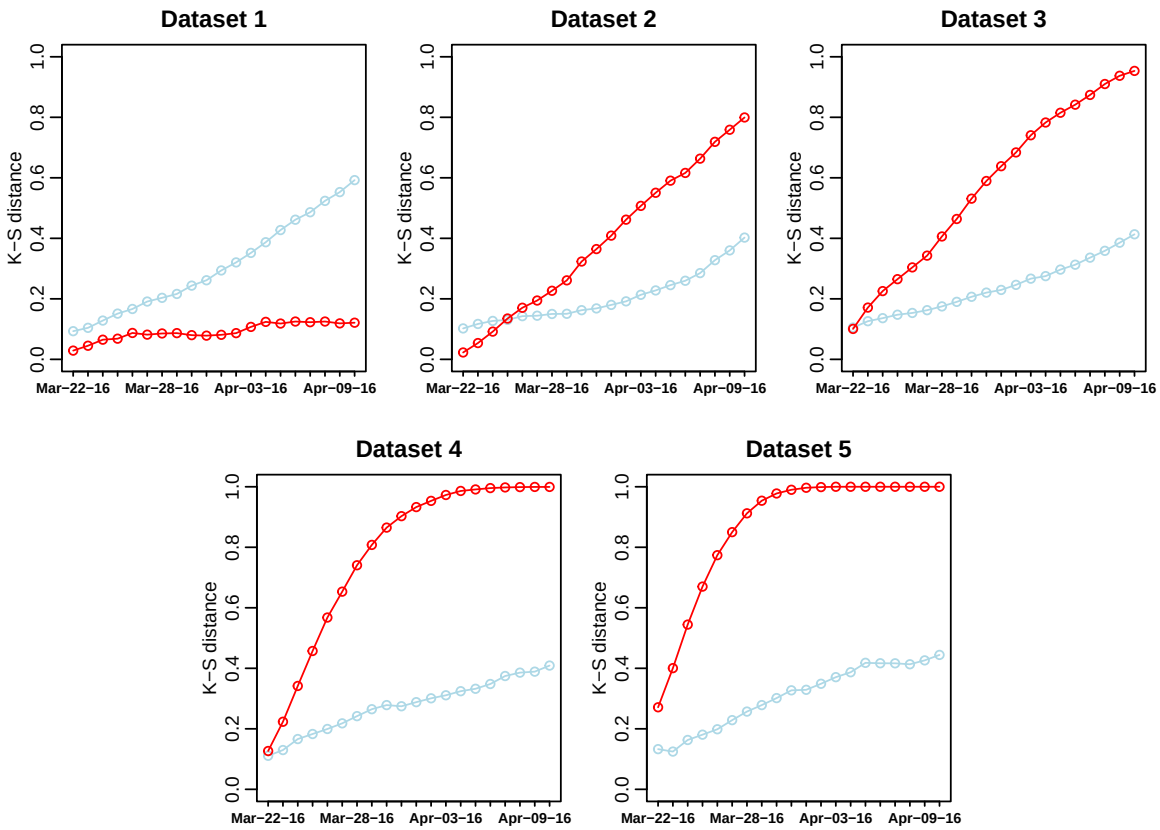

Figure 5: Results of applying the new method to detect causal impacts in Dataset 1 to Dataset 5 using the multivariate model with a stationary local linear trend during the causal period from March 22, 2016 to April 9, 2016. In each subplot, the red line represents the one-sided Kolmogorov-Smirnov (KS) distances between two posterior distributions, one of which is computed using counterfactual data; the light blue line represents the corresponding thresholds.

Date	**Metric**	**Dataset 1**	**Dataset 2**	**Dataset 3**	**Dataset 4**	**Dataset 5**
March 22 (1st day)	KS distance	0.004	0.031	0.100	0.126	0.271
	Threshold	0.108	0.093	0.112	0.110	0.130
March 31 (10th day)	KS distance	0.121	0.391	0.583	0.865	0.977
	Threshold	0.307	0.162	0.206	0.278	0.309
April 9 (20th day)	KS distance	0.514	0.799	0.953	0.999	1.000
	Threshold	0.704	0.368	0.413	0.409	0.616

Table 1: The table shows the one-sided KS distances and corresponding thresholds obtained using the new method for detecting causal impacts across Dataset 1 to Dataset 5. These results are based on a multivariate model incorporating a stationary local linear trend. We report values specifically for March 22, 2016; March 31, 2016; and April 9, 2016, which represent the 1st, 10th, and 20th days of the causal period, respectively.

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